Akhil Reddy Donakonda

Full Stack Developer | Node.js, React.js, AI & Voice Platforms

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PROFESSIONAL SUMMARY

Full Stack Developer with nearly 6 years of experience building scalable SaaS products and distributed systems in cloud telephony, fintech and insurance, plus an internship on **Amazon's FinTech team**. Strong in Node.js, React.js, MongoDB, Redis and RabbitMQ, with production work in event-driven microservices, real-time systems (WebSockets, WebRTC, Asterisk) and LLM-powered features such as voice bots and call quality analysis. Designed the platform's event-driven post-call architecture, scaled a predictive dialer from **10K to 500K calls per day**, and often acts as product owner, taking features from idea to production.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, Python, SQL
Frontend: React.js, Redux, Material UI, Socket.IO, WebRTC, HTML5, CSS3, AngularJS
Backend: Node.js, Express.js, REST APIs, Microservices, WebSockets
Architecture: System Design, Distributed Systems, Event-Driven Architecture, Message Queues, Idempotency, Observability
Databases & Messaging: MongoDB (aggregation, indexing), Redis, MySQL, Oracle, SQL Server, RabbitMQ, Webhooks, Cron
AI/ML: Large Language Models (LLMs), OpenAI GPT-4o, Whisper, Claude API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Speech-to-Text, Text-to-Speech
Voice & Telephony: Asterisk ARI, VoIP, IVR, Predictive Dialer, Voice Bots
Cloud & DevOps: AWS (EC2, S3, Lambda), Google Cloud Platform (GCP), Docker, CI/CD, Bitbucket, Git, Linux, Jira

PROFESSIONAL EXPERIENCE

Full Stack Developer | Matr Innovations (formerly Ten20 Infomedia), Hyderabad Oct 2022 - Present

*Core developer on Doosra and Alohaa, cloud telephony SaaS platforms used by 50+ businesses and 1,000+ agents, handling **1.1M+ calls per day** (answered and unanswered) across predictive dialing, voice APIs and voice streaming.*

- **Designed the event-driven post-call architecture** consolidating 5 call flows (outbound, inbound, predictive, click-to-call, voice bot) onto one RabbitMQ topic exchange (~5M messages per day), isolating fast lanes (billing, metrics) from slow lanes (S3 recording, AI analysis, customer webhooks) so the dialer never waits on a consumer; **shipped the first flow and led the team** through migrating the rest, mentoring 3 junior developers.
- Made billing safe under message redelivery with idempotent handlers and atomic credit deduction, replacing billing logic duplicated across 5 services; added per-queue retries via a delayed exchange and a failed-events store for replay, eliminating silent event loss.
- Designed and built a **predictive dialer** from scratch, scaling it from **10K to 500K calls per day (750K peak)**, with per-tenant rate and concurrency limits in Redis to protect shared calling APIs.
- **Conceived and built the reports product end to end, solo**, from requirements to React UI and Node.js backend: call and agent productivity reports with filters, async generation of large exports via RabbitMQ workers, live progress tracking, presigned S3 downloads and email delivery.
- Streamed exports from MongoDB cursor to CSV, gzip and S3 with Node.js backpressure, fixing out-of-memory crashes for 350K+ calls/day tenants and keeping **peak memory flat at ~350 MB**; cut dialer latency report load from **10-30 s to milliseconds** by precomputing p50/p90/p99 metrics every 5 minutes.
- Built real-time **AI voice bots** using a streaming speech-to-text (STT), LLM and text-to-speech (TTS) pipeline, connecting Asterisk PBX to bots over WebSockets with barge-in; voice streaming handles **100K+ calls per day (200K peak)**.
- Built an **AI call quality analysis** module: Whisper transcription of every call with GPT-4o summaries, sentiment and Q&A.
- Developed **React.js** features across the agent dashboard and WebRTC softphone, AI QA analytics dashboards, campaign management UI and admin console, using Redux, reusable Material UI components, lazy loading, memoization and virtualized lists; replaced UI polling with Socket.IO push updates, **cutting API calls by 30%**.
- Built a voice and SMS broadcast platform (4 microservices) and a programmable voice API handling **500K calls per day (750K peak)** with webhook callbacks and IVR analytics.
- Built HubSpot and Zoho CRM integrations, an in-house OTP classifier replacing a paid service, and production monitoring.

Associate Product Developer | EdgeVerve Systems (Infosys) Oct 2021 - Oct 2022

- Developed and enhanced REST APIs in Java for **Finacle**, Infosys's core banking platform used by banks worldwide; built product features using AngularJS, Java and Oracle SQL, and resolved production defects.

Associate Professional Software Engineer | DXC Technology Jan 2021 - Oct 2021

- Built full-stack features for **Swiss Re**, a global reinsurer, using Java, JSP and SQL Server in Agile Scrum teams.

Software Development Engineer Intern | Amazon Jun 2019 - Sep 2019

- Worked on Amazon's **FinTech** team, building internal tools that solve finance problems across Amazon's businesses, using **Python** and **React.js**.
- Built Python-based data processing on **AWS** services to manage **large volumes of financial data** efficiently.

PROJECTS

- **Employee Background Verification Platform (Freelance) | React.js, Node.js, MongoDB:** Digitized employee background verification for a client, **cutting onboarding time by 40%**.
- **PDF Q&A Assistant | RAG, LLM, Embeddings, Vector Search:** Chunks and embeds uploaded PDFs, retrieves relevant passages per question and sends them to an LLM for grounded answers with conversation context.

EDUCATION

B.Tech, Computer Science and Engineering | VNR Vignana Jyothi Institute of Engineering and Technology CGPA 8.7 | 2016 - 2020